A NORMAL RING IS INTEGRALLY CLOSED IN ITS TOTAL RING OF FRACTIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Normality checked after localising at primes. Used in the counted proof at lines 8442-8445. Reduces the integral element to a fraction over each local ring, where normality is the hypothesis available.
- Flatness of localisation for the total ring of fractions. Used at proof line 8443. Places the total ring of fractions inside its localisation so that the fraction representation can be compared.

Lemma 0.1. *A normal ring is integrally closed in its total ring of fractions.*

Proof. Let R be a normal ring. Let $x \in Q(R)$ be an element of the total ring of fractions of R integral over R. Set $I = \{f \in R, fx \in R\}$. Let $\mathfrak{p} \subset R$ be a prime. As $R \to R_\mathfrak{p}$ is flat we see that $R_\mathfrak{p} \subset Q(R) \otimes_R R_\mathfrak{p}$. As $R_\mathfrak{p}$ is a normal domain we see that $x \otimes 1$ is an element of $R_\mathfrak{p}$. Hence we can find $a, f \in R$, $f \notin \mathfrak{p}$ such that $x \otimes 1 = a \otimes 1/f$. This means that $fx - a$ maps to zero in $Q(R) \otimes_R R_\mathfrak{p} = Q(R)_\mathfrak{p}$, which in turn means that there exists an $f' \in R$, $f' \notin \mathfrak{p}$ such that $f'fx = f'a$ in R. In other words, $ff' \in I$. Thus I is an ideal which isn't contained in any of the prime ideals of R, i.e., $I = R$ and $x \in R$. □

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